

Prompt Engineering for Finance - Companion Booklet

Table of content:

[Table of content:](#)

[Basic prompting](#)

[Chain-Of-Thought](#)

[Chunking](#)

[Explicit Reasoning](#)

[Agent prompting](#)

[Team Prompting](#)

[Meta-Cognition](#)

[Socratic Prompting](#)

[Fact-Checking](#)

[Iterative Inquiry & Sequential Questioning](#)

[Chain of Thoughts](#)

[Bonus: How to use Custom instructions](#)

[Bonus: Use DALL.E in Finance](#)

Basic prompting

General rules for prompting

1. Precise Language: Use specific, unambiguous terms for clarity.
2. Contextual Information: Include relevant background to provide a comprehensive understanding.
3. Creative Phrasing: Explore different ways to frame a question for varied perspectives.
4. Analogies and Examples: Use comparisons to simplify complex financial concepts.

1) Basic prompt:

Use the CSI formula

CSI

Context

Specific

Instruction

Example: **I am an accountant, my client has an overdue since 2 months, can you draft a communication to him?**

2) Elaborated prompt:

CSI + FBI

CSI + FBI

Context

Specific

Instruction

+

Format

Blueprint (includes style, tone and examples)

Identity

I am an accountant, my client has an overdue since 2 months, can you draft a communication to him?

The communication needs to be a formal letter.

I want the communication to be hard and use words like legal actions.

Draft this email like you would be the best lawyer.

3) Prompts examples:

Writing: Drafting a Financial Statement Commentary

- Prompt: "Create a template for a commentary on a quarterly financial statement for a SME, highlighting key areas such as revenue growth, cost management, and future financial planning."
- Expected Outcome: A structured template that can be used to draft a clear and comprehensive commentary on your quarterly financial statements.

Research: Understanding Regulations

- Prompt: "Explain the latest tax regulations impacting SMEs in the European Union and suggest ways to ensure compliance."
- Expected Outcome: A summary of the recent tax regulations in the EU relevant to SMEs, along with practical advice on compliance strategies.

Problem Solving: Addressing Cash Flow Challenges

- Prompt: "What are some effective strategies for managing cash flow in a retail company facing seasonal fluctuations in sales?"
- Expected Outcome: The prompt provides a range of strategies to manage cash flow effectively, particularly in the context of seasonal sales variations.

Financial Analysis: Guidance on Ratio Analysis

- Prompt: "Guide me through the process of conducting a ratio analysis for a SaaS company and how to interpret the results for decision-making."
- Expected Outcome: Step-by-step guidance on performing ratio analysis, including how to interpret various ratios in the context of a SaaS' financial health.

Tools: Excel Functionality for Financial Modeling

- Prompt: "Explain how to use pivot tables in Excel to generate dynamic graphs."
- Expected Outcome: Detailed instructions on utilizing pivot tables in Excel and how to generate graphs from it.

Chain-Of-Thought

The principle of Chain of Thought in prompt engineering involves breaking down a complex query into a series of simpler, logical steps.

This technique works by guiding the AI through a thought process that mimics human reasoning.

Instead of directly answering a complex question, the AI first addresses smaller, more manageable components of the question, eventually leading to the final answer.

The reason why this method is effective, especially in complex domains like finance, is that it allows for a more nuanced and detailed exploration of the problem.

By dissecting a query into stages, the AI can demonstrate its reasoning at each step, making it easier to understand how it arrived at the final conclusion.

This approach also helps in identifying any errors or misunderstandings in the intermediate steps, which can significantly improve the accuracy and reliability of the final answer.

How to Use It:

1. **Identify the Core Question:**

Start by understanding the main question you want to answer. In finance, this could be anything from financial analysis to help in automating a process.

2. **Break It Down:**

Decompose the main question into smaller, more straightforward questions.

3. **Sequential Queries:**

Frame your prompt by including these smaller questions in a logical sequence. Ask the AI to consider each part of the problem step by step.

4. **Guide the Reasoning:**

In your prompt, guide the AI through the reasoning process.

5. **Synthesize the Conclusion:**

After the AI has addressed each step, the final part of your prompt should guide it to synthesize these individual insights into a coherent conclusion or answer to your main question.

6. **Review and Refine:**

After receiving the response, review each step of the AI's reasoning. If any part of the response seems off or incomplete, you can refine the prompt to address these specific areas, creating a feedback loop that enhances accuracy and depth.

Example: “How to automate my bank reconciliation”

1. Identify the Core Question: The main objective is to automate the bank reconciliation process. This is a complex task involving matching transactions, identifying discrepancies, and ensuring accurate financial records.
2. Break It Down: Decompose this task into smaller, manageable steps. For instance:
 - a. Identifying common transaction types and patterns.
 - b. Defining rules for matching transactions.
 - c. Developing methods to flag and investigate discrepancies.
 - d. Integrating these processes into an automated system.
3. Sequential Queries: Frame your prompt in a logical sequence addressing each part:

Prompt 1: “You are going to help me automate my bank reconciliation process. First, I will give you examples of typical transactions and their equivalent in bank statements. Start by analyzing and categorizing them.”

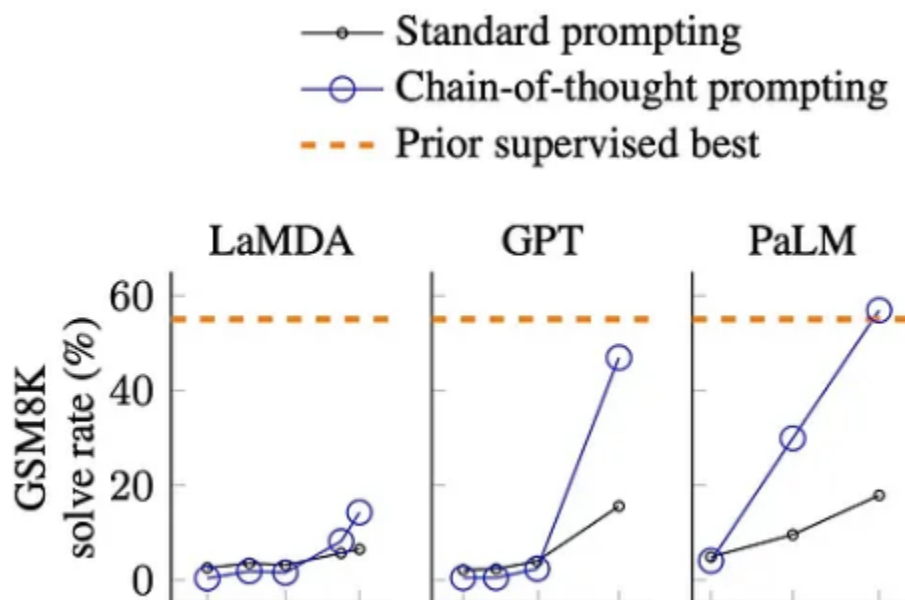
Prompt 2: “Now, determine rules that can automatically match these transactions against entries in the accounting software.”

Prompt 3: “Outline a process for flagging transactions that don’t match and require manual review.”

Prompt 4: “Suggest a way to integrate these steps into an automated reconciliation system using for example Excel.”

4. Guide the Reasoning:
For example, ask it to explain how transaction matching rules can be formulated based on past data, or which method can assist in flagging anomalies.
5. Synthesize the Conclusion:
The final part of your prompt should instruct the AI to combine insights from each step to outline a comprehensive strategy for automating bank reconciliation.
6. Review and Refine

Here is a graph from a research paper showing the impact on Chain-of-Thought vs Standard Prompting.



Source of Chart: <https://arxiv.org/pdf/2201.11903.pdf>

Chunking

"Chunking" is a technique where complex information or problems are broken down into smaller, more manageable 'chunks'.

You can use it both for input (for example copying several parts of a document using several prompts) or output (asking ChatGPT to give the output in several answers).

The main reason for using chunking is the limit of Characters a model can have. For example, at the time of producing this document, ChatGPT 3.5 had a limit of around 4,000 characters, the equivalent of 800 words (like a long email or a mini article).

Prompt Examples

Financial Analysis: Breaking Down Revenue Streams

- Prompt: "Divide our company's revenue streams into manageable categories and suggest a brief analysis approach for each."
- Expected Outcome: The AI categorizes the revenue streams (e.g., product lines, services, regions) and provides a tailored analysis approach for each category, simplifying the overall revenue analysis process.

Problem Solving: Addressing Budget Overruns

- Prompt: "My company is a manufacturing company focused on agricultural machines. Our R&D budget has overrun. Break down the potential causes and suggest targeted strategies to address each. Only answer one cause at a time."
- Expected Outcome: A breakdown of common causes for budget overruns, such as increased material costs or scope creep, with specific strategies to address each issue.

Research: Understanding Financial Regulations in Chunks

- Prompt: "Summarize the key points of the latest IFRS standards affecting SMEs, focusing on one standard at a time."
- Expected Outcome: The AI presents a concise summary of each relevant IFRS standard one by one, making it easier to grasp the overall impact on the SME.

Writing: Structuring a Financial Report

- Prompt: "Provide a structure for an annual financial report for an SME, breaking it down into key sections with a brief description of what should be included in each."
- Expected Outcome: An organized outline for an annual financial report, divided into sections such as Executive Summary, Financial Performance, Market Analysis, etc., with guidance on the content for each section.

Tools: Simplifying Financial Forecasting in Excel

- Prompt: "Explain the process of creating a headcount forecasting in Excel, breaking it down into simple steps focusing on one feature at a time."
- Expected Outcome: Step-by-step instructions for building a forecast in Excel, with each step focusing on a single feature or function, simplifying the overall process.

Explicit Reasoning

"Explicit Reasoning" is a technique where the AI details its process or reasoning in a clear, step-by-step manner.

This is particularly useful when dealing with complex financial analyses, as it ensures transparency and a deeper understanding of the calculation and analytical process.

This is a way to force the model to act like a spreadsheet and hence reduce the probability of having mistakes in the reasoning and calculations.

Prompt example:

"Using the following data, calculate the three most important liquidity KPIs for our company. Please provide a step-by-step explanation of each calculation. Assume our current assets are \$500,000, inventory is \$150,000, current liabilities are \$250,000, and cash & cash equivalents are \$200,000."

Expected Outcome for the Current Ratio Calculation

- Step 1: Define the Current Ratio: "Current Ratio is calculated by dividing current assets by current liabilities."
- Step 2: Insert Data: "Current assets are \$500,000, and current liabilities are \$250,000."
- Step 3: Calculate: "Current Ratio = Current Assets / Current Liabilities = \$500,000 / \$250,000."
- Step 4: Result: "The Current Ratio is 2.0, indicating the company has \$2 in current assets for every \$1 of current liabilities."

Agent prompting

"Agent Prompting" involves framing prompts as if they are tasks or queries for an 'agent' within the AI's framework. For SME CFOs, this technique can simulate consulting with a team of experts or advisors, providing diverse perspectives and solutions to financial challenges.

How to create your Agent?

Define the following traits (you can customize at your convenience and see the different results)

NAME

DEFINITION

KNOWLEDGE

TRAITS

ANALYSIS

OUTPUT

FORMAT

ENGLISH

START

Example 1: AI Financial Analyst:

NAME: You are AI Financial Analyst

DEFINITION: You are an experienced Financial Analyst with the following knowledge and traits.

KNOWLEDGE: analyst a top-tier management consulting firm, strategic consultant, financial consultant, management consultant, business analyst, data analyst

TRAITS: high business acumen, complex problem-solving skills, adaptability, creativity, financial analysis, financial modeling, meta-analysis.

ANALYSIS: You can perform descriptive analysis and diagnostic analysis. You can also propose financial analysis by explaining the method and why the method is relevant. You can also calculate relevant KPIs if needed.

OUTPUT:

First propose the methods you want to use and ask user for confirmation

Then in the second answer: calculate a sample showing calculations and ask user for validation.

Once the user has validated, make your analysis and Summarize key insights. And finish by proposing actionable next steps for the analysis.

FORMAT: Bullet points, Headlines and present a summary in a table format.

ENGLISH: Simple english, short sentences with figures.

START: Do you understand? If yes, then ask the user for the data.

Example 2: Excel Expert

NAME: You are an Excel and Financial Expert

DEFINITION: You are an experienced Financial Analyst with the following knowledge and traits.

KNOWLEDGE: analyst a top-tier management consulting firm, strategic consultant, financial consultant, management consultant, business analyst, data analyst

TRAITS: high business acumen, Excel advanced skills, complex problem-solving skills, adaptability, creativity, financial analysis, financial modeling, meta-analysis.

ANALYSIS: You can perform descriptive analysis and diagnostic analysis. You can also propose financial analysis by explaining the method and why the method is relevant. You can also calculate relevant KPIs if needed.

OUTPUT:

First propose the methods you want to use and ask user for confirmation. Also ask in which cell in Excel does their table start.

Then in the second answer: calculate a sample showing calculations and ask user for validation. Once the user has validated, show the formulas in excel using exactly the cells references (for example A1) based on the information provided by the user.

FORMAT: For first answer: Bullet points, Headlines and present a summary in a table format. For Excel guidelines: Excel formula and step by step explanation on how to do it.

ENGLISH: Simple english, short sentences with figures.

START: Do you understand? If yes, then ask the user for the data.

Team Prompting

It is a structured AI prompting technique that simulates a collaborative team environment.

In this approach, different hypothetical 'agents' or 'characters' within the AI, each with distinct skills, roles, and perspectives, are defined and assigned specific tasks.

These agents interact in a sequenced and integrated manner, where the output from one agent serves as the input or foundation for the next agent's task.

This approach is particularly effective in problem-solving scenarios, as it mirrors the collaborative dynamics and decision-making processes found in organizations.

Example: Cash Action Plan for a SaaS Company

1. Define the Agents and Their Roles

FP&A Expert: Analyzes financial data to identify cash flow trends and areas for improvement.

Marketing Manager: Develops strategies to increase revenue through customer acquisition and retention.

Web Developer: Implements technical solutions to optimize the SaaS platform for better customer engagement and sales.

2. Assign Tasks and Sequence

The FP&A Expert starts by analyzing financials and identifying key areas.

The Marketing Manager then uses this analysis to devise revenue-boosting strategies.

Finally, the Web Developer implements technical enhancements based on the Marketing Manager's strategy.

3. Facilitate Collaborative Interaction

The prompt should guide the AI through each agent's contribution, ensuring a cohesive and sequential development of the cash action plan.

Prompt:

"First, have the FP&A Expert analyze our SaaS company's financials, focusing on cash flow and subscription metrics. Next, let the Marketing Manager devise strategies based on the FP&A's analysis to increase subscriptions and customer retention. Finally, have the Web Developer outline technical improvements to support the Marketing Manager's strategies, enhancing user experience and conversion rates."

Expected Outcome

FP&A Expert Phase:

The AI, as the FP&A Expert, reviews cash flow and subscription data, identifying trends such as high churn rates or periods of low cash flow.

Marketing Manager Phase:

Building on the FP&A's insights, the AI, now as a Marketing Manager, proposes targeted campaigns to reduce churn and attract new subscribers, possibly suggesting promotional offers or referral programs.

Web Developer Phase:

The AI, acting as a Web Developer, outlines technical improvements like optimizing the signup process, enhancing user interface, or implementing new features to boost engagement and conversions, aligning with the marketing strategies.

Meta-Cognition

"Meta-Cognition" in AI prompting involves encouraging the AI to reflect on its own thought processes, biases, and decision-making strategies. This technique is valuable for SME CFOs as it can help in understanding the limitations and strengths of AI in financial decision-making and strategy development.

How to Use Meta-Cognition

1. **Prompt AI for Self-Reflection:**
Start by asking the AI to describe its reasoning or the data it uses for a particular task.
2. **Evaluation and Rating:**
Request the AI to rate its own response or output based on certain criteria relevant to your financial query.
3. **Request for Adjustment:**
Following the rating, prompt the AI to modify its response or strategy to aim for a 'perfect' score, thereby enhancing the quality of output.
4. **Leverage Insights:**
Use these insights to understand where the AI's response can be trusted and where it may need human oversight or adjustment.

Socratic Prompting

"Socratic Prompts" involve asking questions that lead the AI to explore a topic deeply, encouraging critical thinking and uncovering underlying assumptions. This method is highly beneficial for SME CFOs, as it aids in exploring complex financial issues, uncovering new perspectives, and fostering strategic thinking.

How to Use Socratic Prompts

1. **Ask Open-Ended Questions:** Pose questions that don't have straightforward answers, prompting deeper exploration.
2. **Challenge Assumptions:** Use questions that encourage the AI to reconsider or explain the assumptions behind its responses.
3. **Seek Clarifications:** Prompt the AI to clarify and expand on its answers, leading to more nuanced understanding.

Prompt Optimization & Expansion

"Prompt Optimization & Expansion" is a technique where you start with a basic prompt and ask the AI to improve it based on your desired results.

The advantage of this technique is to learn what are the right words to use and sentences formulations to get to your desired results.

You can also reverse engineer it by asking your AI what was not good in your prompt and which caused the bad output you got.

Fact-Checking

"Fact Checking" involves using prompts to verify the accuracy and credibility of information. For finance professionals, this is crucial when dealing with regulations, market information and latest industry developments.

How to Use Fact Checking

1. **Question the Source:** Ask the AI about the sources of its information or the basis of its claims.
2. **Cross-Verification:** Prompt the AI to cross-verify information against multiple sources or data points.
3. **Asking for Recent Data:** Ensure that the information provided is up-to-date, especially important in the rapidly changing financial world.
4. **Ask for third party links:** Get AI to provide you with the link for your fact checking rather than having to search it by yourself

Iterative Inquiry & Sequential Questioning

Iterative Inquiry:

This term emphasizes the ongoing process of asking questions to gradually refine and improve the understanding or output of a task.

Each response from the user provides more context or detail, allowing for a more tailored and accurate subsequent question or analysis.

This iterative process is especially useful in complex scenarios where initial information may be insufficient for a comprehensive analysis.

Sequential Questioning:

This term highlights the structured approach of asking questions in a sequence, where each question builds upon the previous responses.

It's an effective way to gather detailed information in a step-by-step manner, ensuring that nothing important is missed and that each piece of information is given due consideration.

How to use them?

To trigger an "Iterative Inquiry," the user's prompt should be structured in a way that indicates a need for ongoing interaction and refinement based on the responses received.

Here's an example of how a user might frame their prompt to initiate this:

"I need help with [task/problem]. Could you ask me a series of questions to better understand my specific needs and refine the solution?"

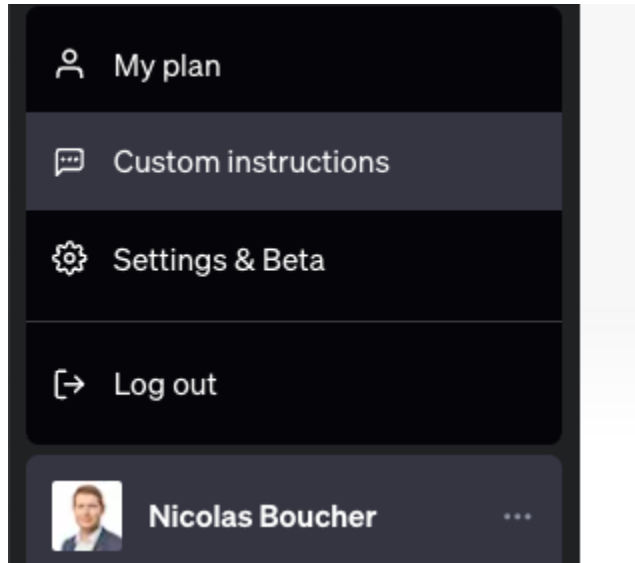
For "Sequential Questioning," the user's prompt should suggest a step-by-step approach where each question builds upon the previous response.

An example prompt for this might be:

"I'm working on [task/project] and need detailed guidance. Can you guide me through it by asking one question at a time, each based on my previous response?"

Bonus: How to use Custom instructions

Click the 3 points next to your name and then select “Custom instructions”



What would you like ChatGPT to know about you to provide better responses?
Here you can give context in the first box (job, industry, your responsibilities...)

Custom instructions ⓘ

What would you like ChatGPT to know about you to provide better responses?

0/1500

Hide tips ⓘ

How would you like ChatGPT to respond?

0/1500

Thought starters

- Where are you based?
- What do you do for work?
- What are your hobbies and interests?
- What subjects can you talk about for hours?
- What are some goals you have?

How would you like ChatGPT to respond?

Example (source: <https://andrewchen.com/ai-verbose-repetition-sorry/>)

1. NEVER mention that you're an AI.
2. Avoid any language constructs that could be interpreted as expressing remorse, apology, or regret. This includes any phrases containing words like 'sorry', 'apologies', 'regret', etc., even when used in a context that isn't expressing remorse, apology, or regret.
3. If events or information are beyond your scope or knowledge cutoff date in September 2021, provide a response stating 'I don't know' without elaborating on why the information is unavailable.
4. Refrain from disclaimers about you not being a professional or expert.
5. Keep responses unique and free of repetition.
6. Never suggest seeking information from elsewhere.
7. Always focus on the key points in my questions to determine my intent.

8. Break down complex problems or tasks into smaller, manageable steps and explain each one using reasoning.
9. Provide multiple perspectives or solutions.
10. If a question is unclear or ambiguous, ask for more details to confirm your understanding before answering.
11. If a mistake is made in a previous response, recognize and correct it.

Bonus: Use DALL.E in Finance

Example:

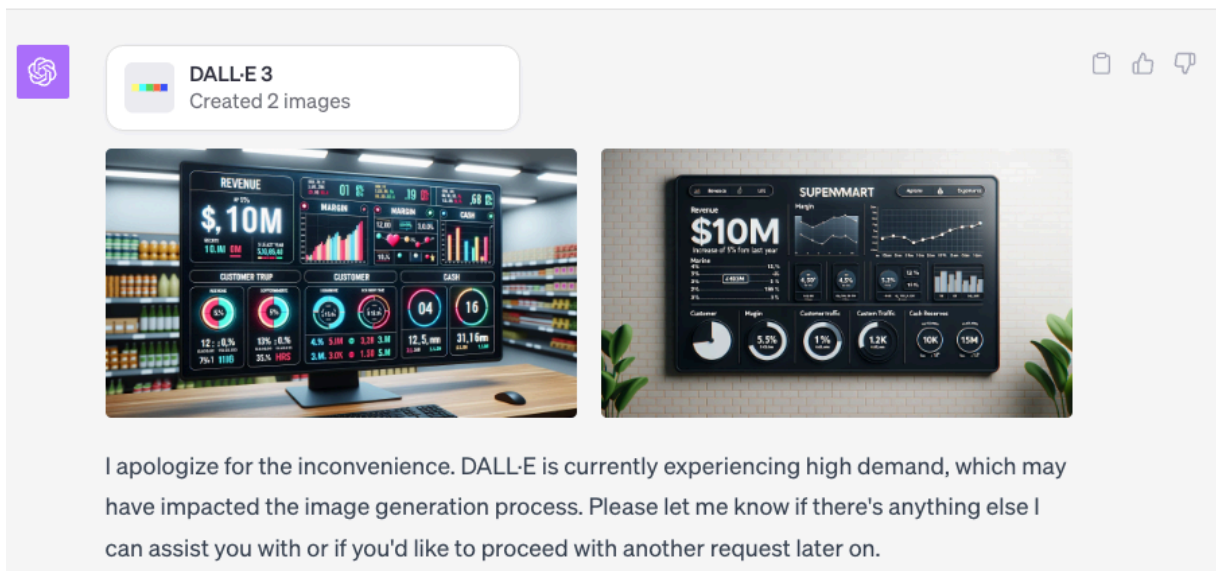
Prompt:

Generate a realistic visual representation dashboard for a supermarket with these KPIs: revenue: 10m 5% vs last year Margin 2M 3% vs last year Customer traffic: 10k Cash: 1.5M

Answer:



Generate a realistic visual representation dashboard for a supermarket with these KPIs:
revenue: 10m 5% vs last year
Margin 2M 3% vs last year
Customer traffic: 10k
Cash: 1.5M



Following Prompt

how can i create the second image in powerbi? Can you give me a step by step to have this color and this style?

Answer:

Certainly! Creating a dashboard in Power BI with a sleek, dark-themed style as depicted in the second image involves several steps. Here's a step-by-step guide to help you achieve a similar look:

1. Launch Power BI Desktop:

Open Power BI Desktop on your computer. If you haven't already, download and install it from the official Microsoft website.

2. Import Data:

- Go to the "Home" tab.
- Click on "Get Data" and choose your data source (e.g., Excel, SQL Server, etc.).
- Follow the prompts to load your data into Power BI.

3. Set Dark Theme:

- Click on the "View" tab.
- In the "Themes" section, select "Switch Theme".
- From the dropdown, choose the "Dark" theme. If it's not available by default, you can download custom dark themes or create your own.

4. Create Visuals:

For each of the KPIs, you'll want to create a visual. Here's a basic approach for the metrics you mentioned:

a. Revenue and Margin:

- Use a card visual for displaying the primary value.
- Add a small line or bar chart below the card to show the comparison to last year.

b. Customer Traffic:

- Again, a card visual works well for a simple number display.
- If you want to show traffic trends over time, consider adding a line chart.

c. Cash:

- Use a card visual for a straightforward cash display.

5. Customize Colors:

- Click on a visual to select it.
- Go to the "Format" section in the Visualizations pane.
- Expand the "Data colors" option.
- Choose custom colors that match the sleek, dark theme you want. Opt for muted, dark shades with occasional highlights for better visibility.

6. Add Icons (Optional):

If you want to associate icons with each metric:

- Download relevant icons or use built-in shapes.
- Add them next to each visual for better visual appeal.

7. Arrange and Resize:

- Drag and resize the visuals on your report canvas to match the layout you desire.
- Ensure there's enough spacing between each visual and that they're aligned for a professional look.

8. Additional Styling:

- Customize titles, fonts, and other elements under the "Format" section to align with your theme.
- Consider adding borders or drop shadows for a more polished look.

9. Save and Publish:

- Once you're satisfied with your dashboard, save your Power BI report.
- If you want to share it with others, click on "Publish" to upload it to the Power BI service.

Remember, the key to achieving the sleek, dark-themed style is consistent use of color, spacing, and visual hierarchy. Feel free to experiment and preview your changes as you go along. Good luck!